Location-Based & Remote Tournament Platform

Comprehensive Design, Legal, and AI Generation Framework

Consolidated Knowledge Transfer Document

Purpose

This document consolidates the design decisions, legal considerations, architecture, and AI workflow developed during the conversation. It is intended as a project context document that can be provided to another LLM or included in a project knowledge base.

Project Vision

The platform is a location-based and online tournament system where players compete in short original games to win real-world prizes.

Unlike a traditional game, the **tournament** is the product.

Individual games exist only as components of a larger tournament architecture.

The platform supports:

- Physical location tournaments
- Remote tournaments
- Mixed tournaments
- Sponsored tournaments
- Brand-sponsored prize pools
- AI-generated games
- AI-generated tournament structures

The long-term goal is to build a system capable of generating thousands of legally defensible tournaments automatically.

Business Model

Revenue sources include:

- SaaS subscriptions for participating businesses
- Advertising
- Sponsored prize categories

- Optional tournament entry fees (carefully structured)
- Anonymous/aggregated analytics
- Enterprise reporting

Businesses pay for:

- Increased foot traffic
- Local engagement
- Analytics
- Promotion

Players receive:

- Entertainment
- Competition
- Rewards
- Prizes

Advertisers receive:

- Audience reach
- Sponsorship opportunities
- Behavioral insights (subject to privacy compliance)

Major Legal Insight

The biggest legal issue is **not** game design.

It is **winner determination**.

Regulators primarily care about:

> What actually determines who receives the prize?

Everything else is secondary.

Lottery Law

Illegal lottery generally consists of:

- Prize
- Chance
- Consideration (payment)

The objective is to ensure tournaments are predominantly determined by skill.

The safest architecture is one where:

- More skilled players consistently outperform less skilled players across repeated tournaments.

Critical Design Principle

The tournament—not an individual game—is the unit of legal analysis.

Originally the assumption was:

"Every game must independently prove skill."

The refined understanding is:

The entire tournament structure should demonstrate skill dominance.

This is a much stronger and more scalable design philosophy.

Tournament Architecture

Tournament

↓

Round 1 → Game A

↓

Round 2 → Game B

↓

Round 3 → Game C

↓

Round 4 → Game D

↓

Winner

Each round is an entirely different game.

Games should test different cognitive abilities.

No single game should unnecessarily determine the tournament outcome unless it is intentionally designed as the strongest skill signal.

Winner Determination Philosophy

Winner determination should satisfy:

- Repeatable
- Explainable
- Skill dominant

The system should always be able to answer:

"Why did Player X win?"

Valid answers include:

- Better strategic decisions
- Higher accuracy
- Better memory
- Better optimization
- Superior reasoning

Invalid answers include:

- Faster tapping
- Better hardware
- Better luck
- Random event

Skill Categories

Games should primarily test:

- Strategy
- Pattern recognition
- Memory

- Planning
- Logic
- Decision-making
- Spatial reasoning
- Resource optimization
- Cognitive flexibility

Mechanical speed should be a supporting factor rather than the dominant determinant.

Weak Mechanics

Examples of weaker mechanics include:

- Pure tapping
- Pure reaction speed
- Repetition
- Hardware-dependent input speed

These mechanics are not automatically prohibited.

However:

- They should not dominate scoring.
- They should not determine winners by themselves.
- They should not be the highest-weighted tournament component.

Tournament Progression

Recommended structure:

Early rounds

Purpose:

- Filter lower-skilled players

Mid rounds

Purpose:

- Differentiate increasingly skilled players

Final rounds

Purpose:

- Demonstrate highest cognitive ability

The strongest skill signals should occur at the end of the tournament.

Controlled Randomness

Randomness is acceptable only if:

- Every player experiences identical conditions.
- Randomness affects presentation rather than competitive advantage.
- Randomness does not determine scoring.

Randomness must never determine who wins.

Device Fairness

Games should minimize:

- Hardware advantages
- Refresh-rate differences
- Input latency
- Platform discrepancies

Suggested methods:

- Server-side validation
- Input normalization
- Anti-cheat systems
- Rate limiting where appropriate

Anti-Cheat Philosophy

Every game should assume malicious users exist.

Detection should include:

- Impossible input frequency
- Automation
- Macros

- Emulators (where appropriate)
- Statistical anomaly detection

Explainability Requirement

Every tournament should be explainable.

An observer should understand why rankings occurred.

The platform should never produce unexplained outcomes.

AI Architecture

The conversation evolved into a layered AI system rather than a single prompt.

The architecture consists of four primary components.

Layer 1 — Dynamic Tournament Builder (DTB)

Purpose:

Design tournament blueprints.

Responsibilities:

- Number of rounds
- Skill progression
- Weighting
- Constraints
- Difficulty curve

Output:

Blueprint only.

No actual games.

Layer 2 — Game Generator

Receives:

- Tournament blueprint
- Framework constraints

Outputs:

Individual game specifications.

Games must satisfy blueprint requirements.

Layer 3 — Compliance Scoring Engine (TCSE)

Purpose:

Evaluate generated tournaments.

Produces:

- Compliance score
- Risk assessment
- Failure conditions
- Suggested revisions

This acts as quality assurance.

Layer 4 — Human Review (Initially)

For MVP:

Human reviews:

- Tournament
- Games
- Scoring
- Compliance

Automation can gradually replace manual review.

AI Pipeline

Tournament Builder

↓

Game Generator

↓

Compliance Engine

↓

Human Approval

↓

Deployment

This is the intended production architecture.

Dynamic Tournament Builder

The DTB does not generate games.

It generates constraints.

Example responsibilities:

- Determine skill domains
- Determine scoring weights
- Determine progression
- Specify prohibited mechanics
- Specify desired cognitive abilities

It creates the blueprint.

Game Generator

Receives one blueprint slot.

Generates:

- One complete game
- Scoring
- Failure conditions
- Rules

It must not modify tournament architecture.

Compliance Engine

Scores:

Skill Dominance

Winner Integrity

Round Structure

Mechanic Quality

Randomness

Device Fairness

Anti-cheat

Repeatability

Automatic rejection occurs when:

- Randomness determines winners
- Weak mechanics dominate
- Final round lacks strong skill signal
- Device advantage determines outcomes

Manual Review Philosophy

Initially:

Every generated tournament should be reviewed manually.

Questions:

Can I explain why the winner won?

Would the same skilled players consistently rank highly?

Is any mechanic legally questionable?

Does any weak mechanic dominate?

Design Philosophy

The objective is NOT merely to make games feel fair.

The objective is to make the tournament:

- Defensible
- Consistent
- Explainable
- Repeatable
- Skill dominant

Practical Development Workflow

Recommended prototype workflow:

1. Generate tournament blueprint.
2. Generate individual games.
3. Evaluate with compliance engine.
4. Manually inspect.
5. Implement.
6. Playtest.
7. Adjust framework.
8. Repeat.

Multi-LLM Workflow

Suggested division of responsibilities:

Reasoning model (Claude, ChatGPT, Gemini, etc.)

Responsibilities:

- Tournament architecture
- Design reasoning
- Blueprint generation
- Compliance analysis

Coding model (Codex, Claude Code, Cursor, etc.)

Responsibilities:

- Game implementation
- Front-end logic
- Server code
- Database code
- APIs

Reviewer model

Responsibilities:

- Independent compliance review
- Design critique
- Risk detection

Long-term goal:

Use specialized models rather than one general-purpose model for every task.

Future Automation

Long-term vision:

Tournament Builder

↓

Game Generator Agents

↓

Automated Compliance Engine

↓

Simulation Engine (optional future enhancement)

↓

Deployment

A future simulation layer could repeatedly simulate tournaments using artificial players of varying skill levels to verify that more skilled simulated players consistently outperform less skilled ones before approving a tournament. This would provide empirical support for the "predominantly skill-based" design goal, though it is not a substitute for legal review.

Privacy & Data Considerations

The discussion identified that user preference selections, gameplay behavior, and location information may qualify as personal information under modern privacy laws even if names are not stored.

Design assumptions:

- Treat behavioral data as regulated.
- Build with transparency.
- Plan for consent and opt-out mechanisms.
- Design aggregation and de-identification carefully before monetizing analytics.

Privacy compliance remains a separate workstream from tournament design.

Physical Event Considerations

For in-person tournaments:

Plan for:

- Participant waivers
- Business agreements
- Insurance
- Venue responsibilities
- Digital acceptance of terms

These are operational requirements outside of the tournament-generation system.

Core Design Principles (Summary)

Every tournament should satisfy these principles:

1. Skill is the dominant determinant of outcome.
2. The tournament, not an individual game, is the primary unit of evaluation.
3. Cognitive skill outweighs pure mechanical execution.
4. Weak mechanics should never determine winners by themselves.
5. Outcomes should be repeatable across multiple runs.
6. Winners should be explainable.
7. Randomness must not determine competitive outcomes.
8. Device differences should not materially affect results.
9. AI should generate within constraints, not invent the constraints.
10. Human review remains essential during MVP development.

Open Questions for Future Research

The following areas require dedicated legal and technical work before production launch:

- State-by-state contest and sweepstakes compliance.
- Paid tournament structures and entry fee analysis.
- Data privacy program (CCPA/CPRA, GDPR, and other applicable laws).
- Advertising disclosure requirements.
- Business agreements with participating venues.
- Insurance requirements.
- AI-generated asset ownership and intellectual property review.
- Tournament simulation and balancing methodology.
- Anti-cheat architecture.
- Accessibility and fairness considerations.

Overall Architectural Philosophy

The key architectural insight from this discussion is to separate responsibilities into distinct layers:

- **Tournament Builder** defines *what* a tournament should test.
- **Game Generator** creates games that satisfy those constraints.
- **Compliance Engine** evaluates whether the resulting tournament remains structurally skill-dominant.
- **Human Review** provides oversight while the platform matures.

This separation reduces design drift, improves consistency, and creates a foundation for scalable tournament generation. It also makes the system easier to audit, iterate, and eventually automate.